IWC-10K

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Industrial High-Power Wireless Charger

Industrial Wireless Power / Autonomous Heavy Vehicles | Rev A1 - Documentation Baseline 28 Jul 2026 | Architecture, Atomic Requirements, Interfaces & Verification

Mission profile. The charger provides zero-contact automated energy transfer at the 10 kW+ class for autonomous mining and port vehicles using resonant inductive coupling. Charging begins only after vehicle identity/permission, pad self-test, alignment/coupling validity and object-detection readiness are established. Detection of metallic debris or live tissue has immediate protection priority over impedance matching, communications and power-request optimization.

Engineering baseline. Ground electronics combine an AC/DC front end, high-frequency SiC inverter, resonant compensation network and drive-over IP68 primary pad. Real-time impedance matching adapts operation across the released coupling window while forced-air or liquid cooling manages ground-pad electronics. Secure Wi-Fi/Bluetooth supports out-of-band vehicle-pad handshake. Manufacturing closes resonance calibration and magnetic-field screening, and the 15-year lifecycle monitors potting degradation under repeated mechanical loading along with electrical and protection health.

Pad geometry, SiC devices, resonant components, FOD/LOD sensing, vehicle receiver interfaces, cooling method, radio profile, potting materials and site mechanical conditions are controlled variants; any change reopens only the affected magnetic, electrical, safety, thermal, mechanical or cybersecurity evidence.

Definition	Controlled product interpretation
Product class	10 kW+ automated resonant-inductive wireless charging system for autonomous mining and port vehicles.
Safety / assurance	Immediate local shutdown on validated metallic-debris or live-tissue detection, with critical electrical/object-protection paths independent of high-level matching and handshake services.
Documentation baseline	28 Jul 2026 - controlled PRD baseline date requested for this product.
Primary architecture	AC/DC front end, high-frequency SiC inverter, resonant network and ground coil, FOD/LOD sensing, Wi-Fi/Bluetooth vehicle-pad handshake, IP68 drive-over ground pad and forced-air or liquid cooling.
Release philosophy	Power-transfer, field/exposure, object detection, dielectric, thermal, mechanical, interoperability and manufacturing evidence are variant-specific release gates.

PRODUCT GOVERNANCE, CHARGING AUTHORITY & REQUIREMENT QUALITY

The wireless charger separates grid-side power conversion, resonant power transfer, vehicle handshake, FOD/LOD protection, thermal/mechanical functions and field-service responsibilities. Charging permission is granted only when local protection prerequisites and authorized vehicle/pad conditions are valid.

High-level charging software cannot overrule a local object-detection or hardware shutdown demand. Governance therefore distinguishes optimization, permission and independent protection authority.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-001	The released configuration shall enforce the following functional allocation: The wireless charger separates grid-side power conversion, resonant power transfer, vehicle handshake, FOD/LOD protection, thermal/mechanical functions and field-service responsibilities.	Controlled architecture, parameter/TBC register, interface matrix, hazard/risk traceability and release evidence map.
IWC10K-002	High-level charging software cannot overrule a local object-detection or hardware shutdown demand. This condition shall trigger the diagnostic or containment behavior allocated by this section instead of silent continuation.	Controlled architecture, parameter/TBC register, interface matrix, hazard/risk traceability and release evidence map.
IWC10K-003	Before accepting a modification to power rating, vehicle interface, pad design, protection mechanism, communication or site installation, the program shall close the affected analysis and Controlled architecture, parameter/TBC register, interface matrix, hazard/risk traceability and release evidence map.	Controlled architecture, parameter/TBC register, interface matrix, hazard/risk traceability and release evidence map.

Verification, margin and future-change rule

Changes affecting power rating, vehicle interface, pad design, protection mechanism, communication or site installation require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

SYSTEM BOUNDARY, GROUND PAD / VEHICLE / SITE OWNERSHIP

The product boundary defines AC/DC supply, ground electronics, primary pad, wireless communication and the declared vehicle-side coil/rectifier interface. Civil works, upstream switchgear, vehicle battery management and traction functions remain external unless allocated.

Unclear ownership of interlocks, grounding, emergency stop or vehicle-ready status can create conflicting charging authority. The ICD assigns each command and feedback to one accountable system.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-004	The product boundary defines AC/DC supply, ground electronics, primary pad, wireless communication and the declared vehicle-side coil/rectifier interface. This behavior shall be preserved for every released variant to which this section applies.	System boundary diagram, site/vehicle ICD and authority table.
IWC10K-005	When the following failure condition occurs - Unclear ownership of interlocks, grounding, emergency stop or vehicle-ready status can create conflicting charging authority - the system shall preserve diagnosability and prevent unqualified normal operation.	System boundary diagram, site/vehicle ICD and authority table.
IWC10K-006	The verification record for this mechanism shall contain System boundary diagram, site/vehicle ICD and authority table.	System boundary diagram, site/vehicle ICD and authority table.

Verification, margin and future-change rule

Reopen this verification topic whenever site supply, ground pad, vehicle receiver, BMS interface, emergency stop or civil installation changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

10 kW+ MISSION, OPERATING MODES & AUTOMATED CHARGING SEQUENCE

The baseline mission is 10 kW+ automated wireless charging for autonomous mining and port vehicles. Modes include de-energized, self-test, vehicle discovery, alignment check, protection validation, low-power characterization, power transfer, controlled stop, fault shutdown and service.

A transition directly to high power without validated coupling, object detection and vehicle permission is prohibited. Restart after a protective trip requires the released re-arm conditions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-007	The product shall satisfy this controlled functional requirement: The baseline mission is 10 kW+ automated wireless charging for autonomous mining and port vehicles.	State-machine test, autonomous docking scenarios, transition interlock test and restart validation.
IWC10K-008	The fault model for this topic includes: A transition directly to high power without validated coupling, object detection and vehicle permission is prohibited. Continued normal authority requires that the relevant validity checks remain satisfied.	State-machine test, autonomous docking scenarios, transition interlock test and restart validation.
IWC10K-009	Changes involving power class, vehicle type, docking sequence, permission policy or service mode shall be assessed against State-machine test, autonomous docking scenarios, transition interlock test and restart validation before the configuration is accepted.	State-machine test, autonomous docking scenarios, transition interlock test and restart validation.

Verification, margin and future-change rule

The change-control trigger here is any modification to power class, vehicle type, docking sequence, permission policy or service mode. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

AC / DC FRONT END, INPUT PROTECTION & ISOLATION BOUNDARY

The ground electronics include the program-defined AC/DC front end, protection and DC bus feeding the high-frequency inverter. Input voltage, power quality, grounding and isolation requirements remain controlled by site architecture.

Grid disturbance or DC-link fault must not prevent local power shutdown. Stored energy and contactor/discharge behavior are included in the fault response.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-010	For all applicable operating states, the mandatory product behavior is: The ground electronics include the program-defined AC/DC front end, protection and DC bus feeding the high-frequency inverter.	Input transient test, protection coordination review, DC-link discharge test and isolation verification.
IWC10K-011	Normal operation shall not be maintained without qualification when this mechanism is present: Grid disturbance or DC-link fault must not prevent local power shutdown.	Input transient test, protection coordination review, DC-link discharge test and isolation verification.
IWC10K-012	The applicable margin and behavior shall be demonstrated through Input transient test, protection coordination review, DC-link discharge test and isolation verification.	Input transient test, protection coordination review, DC-link discharge test and isolation verification.

Verification, margin and future-change rule

For future variants, a change in site voltage, rectifier/PFC, DC-link capacitor, contactor, grounding or upstream protection is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

DC LINK, PRECHARGE / DISCHARGE & STORED-ENERGY CONTROL

DC-link charging and discharge are sequenced so maintenance, fault shutdown and restart occur from known energy states. Voltage sensing supports permissive logic and service diagnostics.

A welded precharge/main contactor or failed discharge path can leave hazardous stored energy after charging stops. Detection and service indication are part of the design.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-013	The implemented function shall preserve this behavior: DC-link charging and discharge are sequenced so maintenance, fault shutdown and restart occur from known energy states.	Precharge timing, welded-contact simulation, discharge-time measurement and voltage-sense plausibility test.
IWC10K-014	The controller shall recognize or bound the following invalid condition: A welded precharge/main contactor or failed discharge path can leave hazardous stored energy after charging stops. Response priority shall follow the mechanism-specific safety or data-validity rule.	Precharge timing, welded-contact simulation, discharge-time measurement and voltage-sense plausibility test.
IWC10K-015	Targeted regression for this topic shall use Precharge timing, welded-contact simulation, discharge-time measurement and voltage-sense plausibility test whenever DC-link capacitance, resistor, contactor, discharge path or voltage sensor changes.	Precharge timing, welded-contact simulation, discharge-time measurement and voltage-sense plausibility test.

Verification, margin and future-change rule

Changes to DC-link capacitance, resistor, contactor, discharge path or voltage sensor are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

HIGH-FREQUENCY SiC INVERTER TOPOLOGY & SWITCHING CONTROL

The primary pad is driven by a high-frequency SiC inverter whose topology, switching frequency, dead time and modulation are selected for the released resonant network and power class. Device ratings and margins remain program-controlled.

Loss of gate control, shoot-through or operation outside the tuned load envelope can create destructive current. Fast electrical protection is independent of optimization software.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-016	This function is part of the released product baseline: The primary pad is driven by a high-frequency SiC inverter whose topology, switching frequency, dead time and modulation are selected for the released resonant network and power class.	Double-pulse/switching characterization, full-load thermal test, shoot-through fault test and modulation verification.
IWC10K-017	This condition is explicitly outside valid normal behavior: Loss of gate control, shoot-through or operation outside the tuned load envelope can create destructive current. The product shall make the loss of validity observable.	Double-pulse/switching characterization, full-load thermal test, shoot-through fault test and modulation verification.
IWC10K-018	The engineering evidence set shall include Double-pulse/switching characterization, full-load thermal test, shoot-through fault test and modulation verification and identify the exact released variant.	Double-pulse/switching characterization, full-load thermal test, shoot-through fault test and modulation verification.

Verification, margin and future-change rule

If SiC module, topology, switching frequency, modulation, DC bus or gate timing changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

SiC GATE DRIVE, DESAT / OVERCURRENT & SAFE TURN-OFF

Gate drivers provide controlled voltage, isolation, dead time and fast fault turn-off appropriate to the selected SiC devices. Protection timing is derived from device fault-energy capability rather than a generic cycle count.

False trips are distinguished from protection failures; desaturation/overcurrent faults latch the power stage safe until the defined reset sequence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-019	The product-level allocation for this topic is: Gate drivers provide controlled voltage, isolation, dead time and fast fault turn-off appropriate to the selected SiC devices.	Gate waveform capture, short-circuit/overcurrent test, propagation-delay budget and reset-latch verification.
IWC10K-020	The design shall prevent silent acceptance of this failure mechanism: False trips are distinguished from protection failures; desaturation/overcurrent faults latch the power stage safe until the defined reset sequence.	Gate waveform capture, short-circuit/overcurrent test, propagation-delay budget and reset-latch verification.
IWC10K-021	Revalidation after changes to gate driver, isolated supply, desat network, gate resistor or SiC device shall close Gate waveform capture, short-circuit/overcurrent test, propagation-delay budget and reset-latch verification.	Gate waveform capture, short-circuit/overcurrent test, propagation-delay budget and reset-latch verification.

Verification, margin and future-change rule

Variant evolution involving gate driver, isolated supply, desat network, gate resistor or SiC device requires a traceable change-impact review that identifies what remains valid and what must be reverified.

RESONANT NETWORK, COMPENSATION & COMPONENT STRESS

The resonant compensation network is tuned for the released coil geometry, coupling range and operating frequency. Capacitor/inductor RMS current, voltage stress and detuning with temperature/aging are included in the design margin.

A component drift can shift current or field strength even when transferred power appears acceptable. Resonant frequency and impedance diagnostics support detection and service.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-022	The controller shall implement the following behavior without transferring authority outside the declared boundary: The resonant compensation network is tuned for the released coil geometry, coupling range and operating frequency.	Impedance sweep, component tolerance analysis, thermal drift test and full-power stress measurement.
IWC10K-023	If this mechanism occurs - A component drift can shift current or field strength even when transferred power appears acceptable - the product shall retain a deterministic and diagnosable response.	Impedance sweep, component tolerance analysis, thermal drift test and full-power stress measurement.
IWC10K-024	The function shall not be considered verified without Impedance sweep, component tolerance analysis, thermal drift test and full-power stress measurement.	Impedance sweep, component tolerance analysis, thermal drift test and full-power stress measurement.

Verification, margin and future-change rule

The verification baseline is reopened when resonant capacitor, inductor, coil, switching frequency or compensation topology changes enough to alter the stated behavior, margin, interface or failure response.

PRIMARY GROUND COIL, MAGNETICS & FIELD SHAPING

The primary coil, ferrite/shield structure and conductor layout produce the required coupling while controlling losses and stray field within the approved installation. Mechanical loads from drive-over service do not alter critical spacing beyond allowed tolerance.

Cracked ferrite, conductor displacement or water ingress can change field distribution and hot spots before gross electrical failure.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-025	The released architecture shall maintain this function across its applicable modes: The primary coil, ferrite/shield structure and conductor layout produce the required coupling while controlling losses and stray field within the approved installation.	Coil resistance/inductance check, field mapping, structural stress correlation and post-drive-over retest.
IWC10K-026	The released fault handling shall cover the following case: Cracked ferrite, conductor displacement or water ingress can change field distribution and hot spots before gross electrical failure. Normal authority shall not persist by assumption alone.	Coil resistance/inductance check, field mapping, structural stress correlation and post-drive-over retest.
IWC10K-027	Evidence reuse after a change to coil geometry, ferrite, shield, conductor, potting or mechanical stack is permitted only where Coil resistance/inductance check, field mapping, structural stress correlation and post-drive-over retest remains representative.	Coil resistance/inductance check, field mapping, structural stress correlation and post-drive-over retest.

Verification, margin and future-change rule

Any modification of coil geometry, ferrite, shield, conductor, potting or mechanical stack is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.

VEHICLE RECEIVER COIL / RECTIFIER INTERFACE BOUNDARY

The ground system defines the electrical and communication contract expected from the vehicle receiver, including compatible frequency/coupling envelope, vehicle-ready state and shutdown acknowledgement. Vehicle-side rectification and battery charging remain owned by the vehicle unless included in the supplied variant.

A receiver with similar nominal power but incompatible tuning or protection behavior is not assumed interoperable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-028	The operational requirement for this mechanism is: The ground system defines the electrical and communication contract expected from the vehicle receiver, including compatible frequency/coupling envelope, vehicle-ready state and shutdown acknowledgement.	Vehicle emulator/receiver compatibility test and interface conformance review.
IWC10K-029	The function shall fail in a controlled manner for this scenario: A receiver with similar nominal power but incompatible tuning or protection behavior is not assumed interoperable.	Vehicle emulator/receiver compatibility test and interface conformance review.
IWC10K-030	Verification of this function shall include: Vehicle emulator/receiver compatibility test and interface conformance review.	Vehicle emulator/receiver compatibility test and interface conformance review.

Verification, margin and future-change rule

Future changes in vehicle coil, rectifier, battery interface, frequency, protocol or vehicle variant are accepted only after the configuration record explains evidence reuse, new risks and the regression needed for closure.

ALIGNMENT SENSING, POSITION WINDOW & DOCKING VALIDATION

Alignment sensing establishes whether lateral, longitudinal and vertical pad position is within the released power-transfer envelope before and during charging. Alignment may use dedicated sensors or inferred coupling metrics, but the validity method is explicit.

A vehicle that shifts during charging can move from efficient coupling into a high-loss or high-field condition. Movement beyond the allowed window initiates derate or shutdown.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-031	The product shall retain the following controlled behavior after startup and during normal operation: Alignment sensing establishes whether lateral, longitudinal and vertical pad position is within the released power-transfer envelope before and during charging.	Alignment sweep, vehicle-motion test, sensor plausibility and boundary-position power test.
IWC10K-032	The design shall explicitly handle this failure mechanism: A vehicle that shifts during charging can move from efficient coupling into a high-loss or high-field condition. The condition shall be diagnosed or bounded before normal authority is retained.	Alignment sweep, vehicle-motion test, sensor plausibility and boundary-position power test.
IWC10K-033	A change to alignment sensor, docking tolerance, vehicle ride height, pad geometry or position algorithm shall trigger impact analysis and targeted regression using Alignment sweep, vehicle-motion test, sensor plausibility and boundary-position power test.	Alignment sweep, vehicle-motion test, sensor plausibility and boundary-position power test.

Verification, margin and future-change rule

Engineering change control for alignment sensor, docking tolerance, vehicle ride height, pad geometry or position algorithm preserves the intent of this page by linking the modified design to updated analysis, test evidence and variant identity.

COUPLING-FACTOR ESTIMATION & OPERATING ENVELOPE

The controller estimates or bounds mutual coupling from approved electrical/sensor observables and uses it to select safe power and matching behavior. The mapping is calibrated to the released pad pair.

An erroneous coupling estimate can command excessive resonant current. Independent electrical limits retain authority over optimization.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-034	The approved implementation shall meet this function-level statement: The controller estimates or bounds mutual coupling from approved electrical/sensor observables and uses it to select safe power and matching behavior.	Coupling fixture sweep, estimator correlation and wrong-calibration negative test.
IWC10K-035	The product shall not accept the following condition as normal valid operation: An erroneous coupling estimate can command excessive resonant current. It shall enter the section-defined invalid, restricted or protective response.	Coupling fixture sweep, estimator correlation and wrong-calibration negative test.
IWC10K-036	Objective release evidence for the stated limits and assumptions shall include Coupling fixture sweep, estimator correlation and wrong-calibration negative test.	Coupling fixture sweep, estimator correlation and wrong-calibration negative test.

Verification, margin and future-change rule

A change to coil pair, estimator, calibration map, gap or alignment range shall not inherit prior approval automatically; the affected evidence is reviewed for continued representativeness before release.

REAL-TIME IMPEDANCE MATCHING & TUNING AUTHORITY

Impedance matching adapts the power stage within released frequency/modulation/component limits to maintain efficient transfer across valid coupling conditions. Matching is an optimization layer and cannot suppress FOD/LOD or overcurrent protection.

Instability or hunting can create thermal cycling and current spikes. Rate limits and validity checks bound adaptation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-037	The released configuration shall enforce the following functional allocation: Impedance matching adapts the power stage within released frequency/modulation/component limits to maintain efficient transfer across valid coupling conditions.	Dynamic misalignment test, tuning step response, stability analysis and protection-priority verification.
IWC10K-038	Instability or hunting can create thermal cycling and current spikes. This condition shall trigger the diagnostic or containment behavior allocated by this section instead of silent continuation.	Dynamic misalignment test, tuning step response, stability analysis and protection-priority verification.
IWC10K-039	Before accepting a modification to matching algorithm, frequency range, modulation limit, control gain or resonant component, the program shall close the affected analysis and Dynamic misalignment test, tuning step response, stability analysis and protection-priority verification.	Dynamic misalignment test, tuning step response, stability analysis and protection-priority verification.

Verification, margin and future-change rule

Changes affecting matching algorithm, frequency range, modulation limit, control gain or resonant component require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

POWER REGULATION, VEHICLE REQUEST & LIMIT ARBITRATION

Transferred power follows the minimum of approved vehicle request, pad capability, thermal limits, coupling/misalignment constraints and local protection limits. The controlled product baseline fixes only the 10 kW+ mission class; exact rated power remains a program-controlled parameter.

A stale vehicle request cannot preserve charging indefinitely. Freshness and communication timeout remove or reduce power authority according to the protocol.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-040	Transferred power follows the minimum of approved vehicle request, pad capability, thermal limits, coupling/misalignment constraints and local protection limits. This behavior shall be preserved for every released variant to which this section applies.	Power-command sweep, limit arbitration test, stale-request timeout and full-load stability verification.
IWC10K-041	When the following failure condition occurs - A stale vehicle request cannot preserve charging indefinitely - the system shall preserve diagnosability and prevent unqualified normal operation.	Power-command sweep, limit arbitration test, stale-request timeout and full-load stability verification.
IWC10K-042	The verification record for this mechanism shall contain Power-command sweep, limit arbitration test, stale-request timeout and full-load stability verification.	Power-command sweep, limit arbitration test, stale-request timeout and full-load stability verification.

Verification, margin and future-change rule

Reopen this verification topic whenever rated power, vehicle request protocol, thermal limit, coupling limit or timeout changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

EFFICIENCY, LOSS BREAKDOWN & THERMAL COUPLING

Efficiency is characterized across the released alignment, gap, power and temperature envelope with separate accounting for inverter, resonant network, coil and auxiliary losses. No universal efficiency number is invented where the baseline does not provide one.

A high system efficiency average can hide localized pad or ferrite hot spots. Loss mapping is correlated to temperature measurements.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-043	The product shall satisfy this controlled functional requirement: Efficiency is characterized across the released alignment, gap, power and temperature envelope with separate accounting for inverter, resonant network, coil and auxiliary losses.	Calibrated input/output power test, calorimetric/thermal correlation as applicable and loss-model review.
IWC10K-044	The fault model for this topic includes: A high system efficiency average can hide localized pad or ferrite hot spots. Continued normal authority requires that the relevant validity checks remain satisfied.	Calibrated input/output power test, calorimetric/thermal correlation as applicable and loss-model review.
IWC10K-045	Changes involving power level, coil geometry, SiC device, resonant component or cooling system shall be assessed against Calibrated input/output power test, calorimetric/thermal correlation as applicable and loss-model review before the configuration is accepted.	Calibrated input/output power test, calorimetric/thermal correlation as applicable and loss-model review.

Verification, margin and future-change rule

The change-control trigger here is any modification to power level, coil geometry, SiC device, resonant component or cooling system. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

FOREIGN OBJECT DETECTION - METALLIC OBJECT SENSING

FOD detects metallic debris or other specified conductive objects within the hazardous region using the released sensing architecture before and during power transfer. Thresholds and test objects are controlled by the product safety plan.

A missed metallic object can heat rapidly; nuisance trips can also impair autonomous operation. Detection coverage, blind spots and response timing are characterized separately.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-046	For all applicable operating states, the mandatory product behavior is: FOD detects metallic debris or other specified conductive objects within the hazardous region using the released sensing architecture before and during power transfer.	Defined-object test matrix, position sweep, thermal correlation and detection-to-shutdown timing.
IWC10K-047	Normal operation shall not be maintained without qualification when this mechanism is present: A missed metallic object can heat rapidly; nuisance trips can also impair autonomous operation.	Defined-object test matrix, position sweep, thermal correlation and detection-to-shutdown timing.
IWC10K-048	The applicable margin and behavior shall be demonstrated through Defined-object test matrix, position sweep, thermal correlation and detection-to-shutdown timing.	Defined-object test matrix, position sweep, thermal correlation and detection-to-shutdown timing.

Verification, margin and future-change rule

For future variants, a change in FOD sensor, algorithm, coil geometry, shield, threshold or test object set is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

LIVE-TISSUE / LIVING OBJECT DETECTION (LOD) & LIMITATIONS

The baseline requires immediate shutdown on detection of live tissue. The released design defines the sensing principle, detection zone, validated object set, limitations and response timing rather than assuming FOD inherently detects people or animals.

Any ambiguity between object classes is resolved conservatively according to the safety concept. Loss of the required LOD channel removes charging permission.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-049	The implemented function shall preserve this behavior: The baseline requires immediate shutdown on detection of live tissue.	LOD test matrix with approved surrogates/methods, coverage mapping and fault injection.
IWC10K-050	The controller shall recognize or bound the following invalid condition: Any ambiguity between object classes is resolved conservatively according to the safety concept. Response priority shall follow the mechanism-specific safety or data-validity rule.	LOD test matrix with approved surrogates/methods, coverage mapping and fault injection.
IWC10K-051	Targeted regression for this topic shall use LOD test matrix with approved surrogates/methods, coverage mapping and fault injection whenever LOD technology, detection zone, threshold, installation geometry or validation method changes.	LOD test matrix with approved surrogates/methods, coverage mapping and fault injection.

Verification, margin and future-change rule

Changes to LOD technology, detection zone, threshold, installation geometry or validation method are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

INDEPENDENT HARDWARE SHUTDOWN PATH & PROTECTION PRIORITY

Critical overcurrent and required object-detection shutdown paths remove gate drive or power authority without depending on high-level matching, cloud service or vehicle convenience software. Priority is hardware/local safety first, optimization second.

A software deadlock must not delay the allocated power-off response. Latched protective shutdown survives until the defined reset conditions are satisfied.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-052	This function is part of the released product baseline: Critical overcurrent and required object-detection shutdown paths remove gate drive or power authority without depending on high-level matching, cloud service or vehicle convenience software.	Fault-path timing capture, software-stall test, gate-inhibit verification and latch/reset test.
IWC10K-053	This condition is explicitly outside valid normal behavior: A software deadlock must not delay the allocated power-off response. The product shall make the loss of validity observable.	Fault-path timing capture, software-stall test, gate-inhibit verification and latch/reset test.
IWC10K-054	The engineering evidence set shall include Fault-path timing capture, software-stall test, gate-inhibit verification and latch/reset test and identify the exact released variant.	Fault-path timing capture, software-stall test, gate-inhibit verification and latch/reset test.

Verification, margin and future-change rule

If shutdown circuit, gate enable, safety logic, contactor or reset policy changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

PROTECTIVE TRIP, RE-ARM & AUTOMATIC RESTART RULES

After FOD/LOD, electrical or thermal trips, the charger enters the appropriate latched or controlled stop state. Automatic restart is allowed only for explicitly classified transient conditions and only after the protection prerequisite is revalidated.

Repeated auto-restart into a persistent object or fault condition is prohibited through attempt limits or latching as defined by the hazard analysis.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-055	The product-level allocation for this topic is: After FOD/LOD, electrical or thermal trips, the charger enters the appropriate latched or controlled stop state.	Trip/re-arm scenario test, persistent-object case, repeated-fault test and operator/service indication review.
IWC10K-056	The design shall prevent silent acceptance of this failure mechanism: Repeated auto-restart into a persistent object or fault condition is prohibited through attempt limits or latching as defined by the hazard analysis.	Trip/re-arm scenario test, persistent-object case, repeated-fault test and operator/service indication review.
IWC10K-057	Revalidation after changes to fault classification, restart policy, attempt limit or service procedure shall close Trip/re-arm scenario test, persistent-object case, repeated-fault test and operator/service indication review.	Trip/re-arm scenario test, persistent-object case, repeated-fault test and operator/service indication review.

Verification, margin and future-change rule

Variant evolution involving fault classification, restart policy, attempt limit or service procedure requires a traceable change-impact review that identifies what remains valid and what must be reverified.

PRIMARY PAD THERMAL SENSING, HOT-SPOT DETECTION & DERATING

Temperature sensing covers power electronics and pad locations needed to protect SiC devices, coils, ferrites, capacitors and potting materials. Derating preserves protection margin across forced-air or liquid-cooled variants.

A single sensor cannot be assumed to represent a large ground pad without thermal correlation. Sensor plausibility and hot-spot coverage are validated by test.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-058	The controller shall implement the following behavior without transferring authority outside the declared boundary: Temperature sensing covers power electronics and pad locations needed to protect SiC devices, coils, ferrites, capacitors and potting materials.	Thermal mapping, sensor correlation, blocked-cooling test and derating/shutdown verification.
IWC10K-059	If this mechanism occurs - A single sensor cannot be assumed to represent a large ground pad without thermal correlation - the product shall retain a deterministic and diagnosable response.	Thermal mapping, sensor correlation, blocked-cooling test and derating/shutdown verification.
IWC10K-060	The function shall not be considered verified without Thermal mapping, sensor correlation, blocked-cooling test and derating/shutdown verification.	Thermal mapping, sensor correlation, blocked-cooling test and derating/shutdown verification.

Verification, margin and future-change rule

The verification baseline is reopened when sensor placement, pad construction, power class, cooling variant or thermal material changes enough to alter the stated behavior, margin, interface or failure response.

FORCED-AIR COOLING VARIANT & CONTAMINATION CONTROL

Forced-air variants control fan health, airflow path, dust/debris exposure and service access for ground electronics. Fan loss results in bounded derating or shutdown based on validated thermal behavior.

Mining/port contamination can reduce airflow long before a fan reports electrical failure. Maintenance indicators and filter/cleaning assumptions are part of the installation plan.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-061	The released architecture shall maintain this function across its applicable modes: Forced-air variants control fan health, airflow path, dust/debris exposure and service access for ground electronics.	Blocked-airflow test, fan fault, contamination inspection and hot-ambient thermal run.
IWC10K-062	The released fault handling shall cover the following case: Mining/port contamination can reduce airflow long before a fan reports electrical failure. Normal authority shall not persist by assumption alone.	Blocked-airflow test, fan fault, contamination inspection and hot-ambient thermal run.
IWC10K-063	Evidence reuse after a change to fan, filter, duct, enclosure opening or site contamination profile is permitted only where Blocked-airflow test, fan fault, contamination inspection and hot-ambient thermal run remains representative.	Blocked-airflow test, fan fault, contamination inspection and hot-ambient thermal run.

Verification, margin and future-change rule

Any modification of fan, filter, duct, enclosure opening or site contamination profile is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.

LIQUID-COOLING VARIANT, LEAK DETECTION & FLOW SUPERVISION

Liquid-cooled variants define coolant interface, flow/temperature sensing, pressure/leak assumptions and electrical isolation from high-frequency power hardware. Loss of cooling removes high-power authority before component limits are exceeded.

A coolant leak can create corrosion or insulation degradation even if temperature remains normal initially. Leak/pressure diagnostics are considered where required by the architecture.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-064	The operational requirement for this mechanism is: Liquid-cooled variants define coolant interface, flow/temperature sensing, pressure/leak assumptions and electrical isolation from high-frequency power hardware.	Flow-loss test, coolant temperature sweep, leak scenario and thermal transient verification.
IWC10K-065	The function shall fail in a controlled manner for this scenario: A coolant leak can create corrosion or insulation degradation even if temperature remains normal initially.	Flow-loss test, coolant temperature sweep, leak scenario and thermal transient verification.
IWC10K-066	Verification of this function shall include: Flow-loss test, coolant temperature sweep, leak scenario and thermal transient verification.	Flow-loss test, coolant temperature sweep, leak scenario and thermal transient verification.

Verification, margin and future-change rule

Future changes in cold plate, pump, coolant, hose, connector or leak-detection method are accepted only after the configuration record explains evidence reuse, new risks and the regression needed for closure.

WI-FI / BLUETOOTH OUT-OF-BAND HANDSHAKE & RADIO COEXISTENCE

The baseline permits secure out-of-band Wi-Fi/Bluetooth for vehicle-pad handshake. The released profile defines discovery, identity, pairing, freshness and charging-permission semantics; radio services do not bypass local protection.

Coexistence with high-power switching and nearby industrial radios is verified so lost packets cause controlled loss of permission rather than stale authorization.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-067	The product shall retain the following controlled behavior after startup and during normal operation: The baseline permits secure out-of-band Wi-Fi/Bluetooth for vehicle-pad handshake.	Radio coexistence test, packet-loss/timeout test, interference scenario and reconnect behavior.
IWC10K-068	The design shall explicitly handle this failure mechanism: Coexistence with high-power switching and nearby industrial radios is verified so lost packets cause controlled loss of permission rather than stale authorization. The condition shall be diagnosed or bounded before normal authority is retained.	Radio coexistence test, packet-loss/timeout test, interference scenario and reconnect behavior.
IWC10K-069	A change to radio module, antenna, channel, protocol profile, timeout or site RF environment shall trigger impact analysis and targeted regression using Radio coexistence test, packet-loss/timeout test, interference scenario and reconnect behavior.	Radio coexistence test, packet-loss/timeout test, interference scenario and reconnect behavior.

Verification, margin and future-change rule

Engineering change control for radio module, antenna, channel, protocol profile, timeout or site RF environment preserves the intent of this page by linking the modified design to updated analysis, test evidence and variant identity.

SECURE PAIRING, VEHICLE / PAD IDENTITY & CREDENTIALS

Vehicle and pad identities are authenticated according to the approved deployment security architecture before charging permission is issued. Credentials are protected against unauthorized extraction or reuse where required by the program.

A previously paired but unauthorized vehicle must not inherit permission solely from proximity or cached state.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-070	The approved implementation shall meet this function-level statement: Vehicle and pad identities are authenticated according to the approved deployment security architecture before charging permission is issued.	Pairing positive/negative tests, credential mismatch/revocation and replay attempt.
IWC10K-071	The product shall not accept the following condition as normal valid operation: A previously paired but unauthorized vehicle must not inherit permission solely from proximity or cached state. It shall enter the section-defined invalid, restricted or protective response.	Pairing positive/negative tests, credential mismatch/revocation and replay attempt.
IWC10K-072	Objective release evidence for the stated limits and assumptions shall include Pairing positive/negative tests, credential mismatch/revocation and replay attempt.	Pairing positive/negative tests, credential mismatch/revocation and replay attempt.

Verification, margin and future-change rule

A change to credential type, pairing method, trust store, vehicle identity or commissioning process shall not inherit prior approval automatically; the affected evidence is reviewed for continued representativeness before release.

CHARGING-PERMISSION STATE MACHINE & AUTHORITY ARBITRATION

Charging permission combines authenticated vehicle readiness, pad self-test, valid alignment/coupling, object-detection readiness, thermal/electrical health and site interlocks. Removing any required permissive causes the specified controlled or immediate stop.

The state machine prevents a communications service from restoring power while a local protective latch remains active.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-073	The released configuration shall enforce the following functional allocation: Charging permission combines authenticated vehicle readiness, pad self-test, valid alignment/coupling, object-detection readiness, thermal/electrical health and site interlocks.	Permission matrix test, simultaneous-fault scenarios and protection-priority verification.
IWC10K-074	The state machine prevents a communications service from restoring power while a local protective latch remains active. This condition shall trigger the diagnostic or containment behavior allocated by this section instead of silent continuation.	Permission matrix test, simultaneous-fault scenarios and protection-priority verification.
IWC10K-075	Before accepting a modification to permissive input, communication protocol, local interlock, object detector or reset logic, the program shall close the affected analysis and Permission matrix test, simultaneous-fault scenarios and protection-priority verification.	Permission matrix test, simultaneous-fault scenarios and protection-priority verification.

Verification, margin and future-change rule

Changes affecting permissive input, communication protocol, local interlock, object detector or reset logic require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

COMMUNICATION LOSS, TIMEOUT & SAFE POWER RESPONSE

Loss or staleness of required vehicle communication removes charging authority within a controlled timeout. The local power stage can always execute protective shutdown without waiting for remote acknowledgement.

Brief packet loss may be tolerated only inside a bounded communication supervision window validated against vehicle and pad behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-076	Loss or staleness of required vehicle communication removes charging authority within a controlled timeout. This behavior shall be preserved for every released variant to which this section applies.	Timeout sweep, link drop during ramp/full power and recovery/re-authentication test.
IWC10K-077	When the following failure condition occurs - Brief packet loss may be tolerated only inside a bounded communication supervision window validated against vehicle and pad behavior - the system shall preserve diagnosability and prevent unqualified normal operation.	Timeout sweep, link drop during ramp/full power and recovery/re-authentication test.
IWC10K-078	The verification record for this mechanism shall contain Timeout sweep, link drop during ramp/full power and recovery/re-authentication test.	Timeout sweep, link drop during ramp/full power and recovery/re-authentication test.

Verification, margin and future-change rule

Reopen this verification topic whenever radio link, timeout, retry policy, vehicle protocol or reconnect behavior changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

CYBERSECURITY, SECURE BOOT / UPDATE & SERVICE ACCESS

Secure boot and authenticated update protect charging control, protection configuration and communication software from unauthorized modification. IEC 62443 or other industrial cybersecurity frameworks are candidates to confirm by program, not automatic compliance claims.

Service access cannot disable FOD/LOD or independent electrical protection outside the authorized engineering process.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-079	The release process shall satisfy this controlled requirement: Secure boot and authenticated update protect charging control, protection configuration and communication software from unauthorized modification.	Secure-boot test, update signature failure, access-control test and security review.
IWC10K-080	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Secure-boot test, update signature failure, access-control test and security review.
IWC10K-081	Release evidence after modifications to bootloader, update mechanism, service port, trust anchor or security standard shall include Secure-boot test, update signature failure, access-control test and security review.	Secure-boot test, update signature failure, access-control test and security review.

Verification, margin and future-change rule

The change-control trigger here is any modification to bootloader, update mechanism, service port, trust anchor or security standard. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

MAGNETIC FIELD EMISSION MAPPING & EXPOSURE BOUNDARY

Magnetic-field mapping characterizes the released pad pair over alignment, gap, power and object conditions. Applicable human-exposure and equipment-interference limits are program/site-controlled because the baseline does not provide numerical field limits.

A local field hot spot may occur away from the measurement positions used for nominal efficiency testing; mapping therefore covers the accessible boundary and credible misalignment.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-082	For each released variant, the following release condition applies: Magnetic-field mapping characterizes the released pad pair over alignment, gap, power and object conditions.	Calibrated field survey, misalignment map and compliance comparison to approved program limits.
IWC10K-083	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Calibrated field survey, misalignment map and compliance comparison to approved program limits.
IWC10K-084	The program shall close Calibrated field survey, misalignment map and compliance comparison to approved program limits before approving the affected production variant.	Calibrated field survey, misalignment map and compliance comparison to approved program limits.

Verification, margin and future-change rule

For future variants, a change in pad geometry, shielding, power, installation, exposure standard or measurement grid is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

EMC, HARMONICS & RADIO / POWER-ELECTRONICS COEXISTENCE

The charger controls conducted/radiated emissions and immunity across the AC/DC stage, high-frequency inverter, pad and radio interfaces. Power switching must not blind FOD/LOD sensors or the handshake radio.

EMC verification monitors charging stability and protection sensor validity during disturbance, not only communication availability.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-085	The configuration shall not be released unless this statement is true: The charger controls conducted/radiated emissions and immunity across the AC/DC stage, high-frequency inverter, pad and radio interfaces.	Emissions/immunity testing, coexistence test and protection-sensor quality monitoring.
IWC10K-086	A missing verification result, manufacturing record or similarity justification for this topic is a release-gate condition and shall not be converted into runtime behavior.	Emissions/immunity testing, coexistence test and protection-sensor quality monitoring.
IWC10K-087	Variant sign-off shall reference Emissions/immunity testing, coexistence test and protection-sensor quality monitoring and the applicable configuration identity.	Emissions/immunity testing, coexistence test and protection-sensor quality monitoring.

Verification, margin and future-change rule

Changes to switching frequency, filter, cable, shield, radio or EMC requirement are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

IP68 GROUND PAD SEALING, DRAINAGE & ENVIRONMENTAL INGRESS

The baseline requires an IP68 drive-over capable ground pad. Sealing, cable exits, pressure equalization if used, drainage and material compatibility are verified for the selected installation; exact immersion conditions follow the program qualification definition.

Water or contaminant ingress can change insulation and resonant behavior before visible failure. Post-ingress electrical, thermal and field tests are therefore required.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-088	Release readiness for this topic depends on the following controlled condition: The baseline requires an IP68 drive-over capable ground pad.	Ingress qualification, insulation/resonance retest and post-exposure functional verification.
IWC10K-089	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	Ingress qualification, insulation/resonance retest and post-exposure functional verification.
IWC10K-090	Any evidence reuse following a change to seal, gland, potting, vent, connector or installation drainage shall be justified against Ingress qualification, insulation/resonance retest and post-exposure functional verification.	Ingress qualification, insulation/resonance retest and post-exposure functional verification.

Verification, margin and future-change rule

If seal, gland, potting, vent, connector or installation drainage changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

DRIVE-OVER STRUCTURE, VEHICLE LOAD & PAD DEFLECTION

The ground pad structure sustains the released vehicle wheel/axle loading, impact and installation support conditions without violating coil spacing, insulation or enclosure integrity. Numerical load classes remain TBC until the vehicle/site are defined.

Permanent deformation can detune coupling or crack ferrites while leaving the enclosure apparently intact. Mechanical qualification therefore includes electrical and field correlation after load testing.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-091	The approved baseline shall demonstrate the following before release: The ground pad structure sustains the released vehicle wheel/axle loading, impact and installation support conditions without violating coil spacing, insulation or enclosure integrity.	Static/dynamic load test, deflection measurement, post-load resonance/field map and inspection.
IWC10K-092	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Static/dynamic load test, deflection measurement, post-load resonance/field map and inspection.
IWC10K-093	The release package shall contain Static/dynamic load test, deflection measurement, post-load resonance/field map and inspection for the exact hardware/software variant.	Static/dynamic load test, deflection measurement, post-load resonance/field map and inspection.

Verification, margin and future-change rule

Variant evolution involving vehicle load, pad frame, support foundation, top plate or coil stack requires a traceable change-impact review that identifies what remains valid and what must be reverified.

POTTING / ENCAPSULATION, DIELECTRIC INTEGRITY & MECHANICAL STRESS

Potting protects coils and electronics from moisture and mechanical stress while maintaining dielectric, thermal and magnetic performance. Material cure, void content, adhesion and thermal expansion are manufacturing-critical characteristics.

Cracks or delamination under repeated drive-over and thermal cycling can create moisture paths and coil movement.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-094	Variant approval is conditional on the following requirement being met: Potting protects coils and electronics from moisture and mechanical stress while maintaining dielectric, thermal and magnetic performance.	Material qualification, void/adhesion inspection, thermal-mechanical cycling and dielectric retest.
IWC10K-095	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Material qualification, void/adhesion inspection, thermal-mechanical cycling and dielectric retest.
IWC10K-096	Production release is contingent on completion of Material qualification, void/adhesion inspection, thermal-mechanical cycling and dielectric retest.	Material qualification, void/adhesion inspection, thermal-mechanical cycling and dielectric retest.

Verification, margin and future-change rule

The verification baseline is reopened when potting material, cure process, thickness, primer, coil construction or supplier changes enough to alter the stated behavior, margin, interface or failure response.

OUTDOOR / MINING / PORT ENVIRONMENTAL QUALIFICATION

Qualification addresses the program-selected temperature, humidity, salt/chemical exposure, dust, vibration, impact and contamination expected at mining or port installations. The baseline does not release universal severity values.

The charger is released only for variants whose environmental matrix and post-stress functional evidence are complete.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-097	This release-gate requirement applies to the exact configuration: Qualification addresses the program-selected temperature, humidity, salt/chemical exposure, dust, vibration, impact and contamination expected at mining or port installations.	Environmental qualification plan, sequential stress and post-stress power/FOD/insulation retest.
IWC10K-098	A missing verification result, manufacturing record or similarity justification for this topic is a release-gate condition and shall not be converted into runtime behavior.	Environmental qualification plan, sequential stress and post-stress power/FOD/insulation retest.
IWC10K-099	Engineering approval after changes to site environment, enclosure, cooling system, material or installation method shall be based on Environmental qualification plan, sequential stress and post-stress power/FOD/insulation retest.	Environmental qualification plan, sequential stress and post-stress power/FOD/insulation retest.

Verification, margin and future-change rule

Any modification of site environment, enclosure, cooling system, material or installation method is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.

MISALIGNMENT, DEBRIS & SURFACE-CONDITION ROBUSTNESS

Operation is verified with credible lateral/longitudinal offset, ride-height variation, mud/water/debris and surface wear while preserving coupling, protection and thermal margins. Conditions outside the validated envelope inhibit charging.

The system does not use matching control to force rated power through a mechanically invalid coupling condition.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-100	Engineering sign-off for this topic shall confirm: Operation is verified with credible lateral/longitudinal offset, ride-height variation, mud/water/debris and surface wear while preserving coupling, protection and thermal margins.	Misalignment/debris matrix, thermal/field measurement and inhibit-threshold verification.
IWC10K-101	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	Misalignment/debris matrix, thermal/field measurement and inhibit-threshold verification.
IWC10K-102	Release approval for this topic shall require: Misalignment/debris matrix, thermal/field measurement and inhibit-threshold verification.	Misalignment/debris matrix, thermal/field measurement and inhibit-threshold verification.

Verification, margin and future-change rule

Future changes in alignment envelope, ride height, surface condition, debris type or power class are accepted only after the configuration record explains evidence reuse, new risks and the regression needed for closure.

EOL RESONANCE CALIBRATION, COIL PARAMETER & IMPEDANCE SCREEN

End-of-line testing measures the released coil/resonant parameters and records calibration values needed by matching and diagnostics. Units outside approved resonance/impedance tolerance are rejected or dispositioned, not software-compensated without engineering approval.

Calibration equipment uncertainty is included in acceptance limits.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-103	The production baseline shall comply with the following release condition: End-of-line testing measures the released coil/resonant parameters and records calibration values needed by matching and diagnostics.	EOL resonance sweep, impedance record, gauge R&R/measurement-system analysis as applicable and serialization.
IWC10K-104	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	EOL resonance sweep, impedance record, gauge R&R/measurement-system analysis as applicable and serialization.
IWC10K-105	A change to coil, capacitor, calibration fixture, measurement method or acceptance limit shall reopen the affected release evidence; closure shall include EOL resonance sweep, impedance record, gauge R&R/measurement-system analysis as applicable and serialization.	EOL resonance sweep, impedance record, gauge R&R/measurement-system analysis as applicable and serialization.

Verification, margin and future-change rule

Engineering change control for coil, capacitor, calibration fixture, measurement method or acceptance limit preserves the intent of this page by linking the modified design to updated analysis, test evidence and variant identity.

EOL MAGNETIC FIELD MAPPING / SENSOR FUNCTION & FOD SCREEN

Manufacturing verifies field-generation/sensing functions and FOD/LOD channel operation using the approved production method. Full design field mapping may be qualification-level, while EOL uses correlated screening criteria.

A unit cannot pass merely because power transfer works if its protection sensing is outside the production acceptance window.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-106	No affected variant may be released until this condition is satisfied: Manufacturing verifies field-generation/sensing functions and FOD/LOD channel operation using the approved production method.	EOL FOD/LOD functional test, field/sensor correlated screen and genealogy record.
IWC10K-107	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	EOL FOD/LOD functional test, field/sensor correlated screen and genealogy record.
IWC10K-108	Production equivalence after a change to production fixture, reference object, sensor, threshold or screening method shall require EOL FOD/LOD functional test, field/sensor correlated screen and genealogy record.	EOL FOD/LOD functional test, field/sensor correlated screen and genealogy record.

Verification, margin and future-change rule

A change to production fixture, reference object, sensor, threshold or screening method shall not inherit prior approval automatically; the affected evidence is reviewed for continued representativeness before release.

MANUFACTURING - COIL GEOMETRY, POTTING, INSULATION & GENEALOGY

Manufacturing controls coil placement, ferrite/shield stack, resonant components, potting process, insulation, cooling interfaces and power-electronics identity as critical-to-quality characteristics.

Process drift that changes coupling or dielectric margin is detected through in-process and EOL controls before release.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-109	Release approval shall enforce the following condition: Manufacturing controls coil placement, ferrite/shield stack, resonant components, potting process, insulation, cooling interfaces and power-electronics identity as critical-to-quality characteristics.	Process control plan, inspection records, HiPot/insulation evidence and serialized genealogy.
IWC10K-110	A missing verification result, manufacturing record or similarity justification for this topic is a release-gate condition and shall not be converted into runtime behavior.	Process control plan, inspection records, HiPot/insulation evidence and serialized genealogy.
IWC10K-111	The released configuration index shall reference the following objective evidence: Process control plan, inspection records, HiPot/insulation evidence and serialized genealogy.	Process control plan, inspection records, HiPot/insulation evidence and serialized genealogy.

Verification, margin and future-change rule

Changes affecting coil process, ferrite, potting, SiC module, cooling assembly or supplier process require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

DVP - POWER TRANSFER, ALIGNMENT, GAP & EFFICIENCY

Full-system DVP measures stable power transfer across approved pad/vehicle variants, alignment, gap, temperature and input conditions with electrical/thermal margin. The baseline 10 kW+ class is preserved while exact release rating is program-controlled.

Missing coverage for a released coupling or power condition blocks release; it is not inferred from a nearby test point.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-112	Full-system DVP measures stable power transfer across approved pad/vehicle variants, alignment, gap, temperature and input conditions with electrical/thermal margin. This condition is mandatory for the affected release.	Calibrated DVP report, power/efficiency maps, temperature data and requirement traceability.
IWC10K-113	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	Calibrated DVP report, power/efficiency maps, temperature data and requirement traceability.
IWC10K-114	This topic remains open in the release matrix until Calibrated DVP report, power/efficiency maps, temperature data and requirement traceability is approved.	Calibrated DVP report, power/efficiency maps, temperature data and requirement traceability.

Verification, margin and future-change rule

Reopen this verification topic whenever power rating, pad pair, gap, alignment, temperature or input supply changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

DVP - FOD / LOD DETECTION & SHUTDOWN TIMING

Protection DVP maps metallic-object and live-tissue detection over the defined hazardous region, including edge positions, misalignment, power levels and sensor faults, while measuring detection-to-gate-inhibit/power-decay timing.

Protection acceptance is tied to the approved hazard analysis and object sets; vague “immediate” wording is converted to a controlled numerical timing requirement before release.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-115	The release process shall satisfy this controlled requirement: Protection DVP maps metallic-object and live-tissue detection over the defined hazardous region, including edge positions, misalignment, power levels and sensor faults, while measuring detection-to-gate-inhibit/power-decay timing.	Object-detection qualification report, timing traces and blind-spot/coverage map.
IWC10K-116	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Object-detection qualification report, timing traces and blind-spot/coverage map.
IWC10K-117	Release evidence after modifications to FOD/LOD sensor, object set, detection zone, timing allocation or pad design shall include Object-detection qualification report, timing traces and blind-spot/coverage map.	Object-detection qualification report, timing traces and blind-spot/coverage map.

Verification, margin and future-change rule

The change-control trigger here is any modification to FOD/LOD sensor, object set, detection zone, timing allocation or pad design. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

DVP - DRIVE-OVER, IMPACT & POST-STRESS ELECTRICAL SAFETY

Mechanical DVP applies the released drive-over/load profile and then repeats insulation, resonance, field, FOD/LOD and thermal checks. Hidden magnetic or dielectric damage is therefore included in acceptance.

Mechanical qualification that stops at visual inspection is insufficient for production release.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-118	For each released variant, the following release condition applies: Mechanical DVP applies the released drive-over/load profile and then repeats insulation, resonance, field, FOD/LOD and thermal checks.	Drive-over/impact report with post-stress electrical and protection retest.
IWC10K-119	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Drive-over/impact report with post-stress electrical and protection retest.
IWC10K-120	The program shall close Drive-over/impact report with post-stress electrical and protection retest before approving the affected production variant.	Drive-over/impact report with post-stress electrical and protection retest.

Verification, margin and future-change rule

For future variants, a change in vehicle load, pad structure, potting, installation support or qualification profile is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

HI-POT / INSULATION, LEAKAGE & ELECTRICAL SAFETY EVIDENCE

Insulation coordination between mains/DC power, high-frequency circuitry, pad structure and accessible/communication circuits is verified according to the selected industrial electrical-safety architecture. Exact test voltages and standards are controlled by the program.

Failed or incomplete dielectric evidence blocks the affected configuration from release.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-121	The configuration shall not be released unless this statement is true: Insulation coordination between mains/DC power, high-frequency circuitry, pad structure and accessible/communication circuits is verified according to the selected industrial electrical-safety architecture.	HiPot/insulation resistance/leakage evidence and design insulation review.
IWC10K-122	A missing verification result, manufacturing record or similarity justification for this topic is a release-gate condition and shall not be converted into runtime behavior.	HiPot/insulation resistance/leakage evidence and design insulation review.
IWC10K-123	Variant sign-off shall reference HiPot/insulation resistance/leakage evidence and design insulation review and the applicable configuration identity.	HiPot/insulation resistance/leakage evidence and design insulation review.

Verification, margin and future-change rule

Changes to isolation barrier, material, spacing, connector, potting or applicable safety standard are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

15-YEAR LIFECYCLE, POTTING DEGRADATION & CONDITION MONITORING

The baseline assigns a 15-year lifecycle with monitoring of potting degradation under heavy mechanical stress. Lifecycle data also tracks coil/resonance drift, insulation indicators, cooling performance, switching-device stress and protection-channel health as supported by the released architecture.

A field indicator is advisory unless correlated to validated degradation limits; maintenance/replacement thresholds are controlled by engineering evidence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-124	Release readiness for this topic depends on the following controlled condition: The baseline assigns a 15-year lifecycle with monitoring of potting degradation under heavy mechanical stress.	Lifecycle model, field-data plan, inspection criteria and replacement-threshold validation.
IWC10K-125	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	Lifecycle model, field-data plan, inspection criteria and replacement-threshold validation.
IWC10K-126	Any evidence reuse following a change to potting material, duty cycle, drive-over profile, thermal cycle, coil drift or maintenance interval shall be justified against Lifecycle model, field-data plan, inspection criteria and replacement-threshold validation.	Lifecycle model, field-data plan, inspection criteria and replacement-threshold validation.

Verification, margin and future-change rule

If potting material, duty cycle, drive-over profile, thermal cycle, coil drift or maintenance interval changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

FIELD DIAGNOSTICS, SERVICE REPLACEMENT & RE-CALIBRATION

Service diagnostics identify active pad/vehicle compatibility, calibration state, protection health, thermal status and fault history. Replacement of coil, resonant component, sensor or power module triggers the required resonance/field/protection recalibration.

A serviced unit does not return to high-power operation until mandatory post-service checks pass.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-127	The approved baseline shall demonstrate the following before release: Service diagnostics identify active pad/vehicle compatibility, calibration state, protection health, thermal status and fault history.	Service procedure validation, replacement simulation and post-service calibration/protection test.
IWC10K-128	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Service procedure validation, replacement simulation and post-service calibration/protection test.
IWC10K-129	The release package shall contain Service procedure validation, replacement simulation and post-service calibration/protection test for the exact hardware/software variant.	Service procedure validation, replacement simulation and post-service calibration/protection test.

Verification, margin and future-change rule

Variant evolution involving field-replaceable part, calibration method, service tool or maintenance workflow requires a traceable change-impact review that identifies what remains valid and what must be reverified.

SUPPLIER CHANGE, SECOND SOURCE & PERFORMANCE EQUIVALENCE

Supplier changes to SiC modules, resonant capacitors, coils/ferrites, FOD/LOD sensors, potting, cooling or radio hardware are assessed for electrical, magnetic, thermal, mechanical and security impact.

Second-source approval requires similarity evidence and targeted regression across the mechanisms actually affected.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-130	Variant approval is conditional on the following requirement being met: Supplier changes to SiC modules, resonant capacitors, coils/ferrites, FOD/LOD sensors, potting, cooling or radio hardware are assessed for electrical, magnetic, thermal, mechanical and security impact.	Supplier change-impact review, similarity analysis and regression DVP.
IWC10K-131	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Supplier change-impact review, similarity analysis and regression DVP.
IWC10K-132	Production release is contingent on completion of Supplier change-impact review, similarity analysis and regression DVP.	Supplier change-impact review, similarity analysis and regression DVP.

Verification, margin and future-change rule

The verification baseline is reopened when critical supplier, material, semiconductor, magnetics, sensor or radio component changes enough to alter the stated behavior, margin, interface or failure response.

FINAL RELEASE CLOSURE & SITE / VEHICLE VARIANT APPROVAL

Release closes only when the selected pad, power stage, receiver interface, protection, communication, mechanical installation and software/calibration combination has complete DVP, manufacturing, safety/security and lifecycle evidence. Site and vehicle variants are explicitly enumerated.

Open FOD/LOD, mechanical, dielectric, interoperability or supplier-equivalence evidence remains a release blocker rather than a fabricated runtime degraded mode.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
IWC10K-133	This release-gate requirement applies to the exact configuration: Release closes only when the selected pad, power stage, receiver interface, protection, communication, mechanical installation and software/calibration combination has complete DVP, manufacturing, safety/security and lifecycle evidence.	Final release checklist, approved variant matrix, deviation log and signed evidence index.
IWC10K-134	A missing verification result, manufacturing record or similarity justification for this topic is a release-gate condition and shall not be converted into runtime behavior.	Final release checklist, approved variant matrix, deviation log and signed evidence index.
IWC10K-135	Engineering approval after changes to any released pad/vehicle/site baseline element or unresolved evidence item shall be based on Final release checklist, approved variant matrix, deviation log and signed evidence index.	Final release checklist, approved variant matrix, deviation log and signed evidence index.

Verification, margin and future-change rule

Any modification of any released pad/vehicle/site baseline element or unresolved evidence item is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.